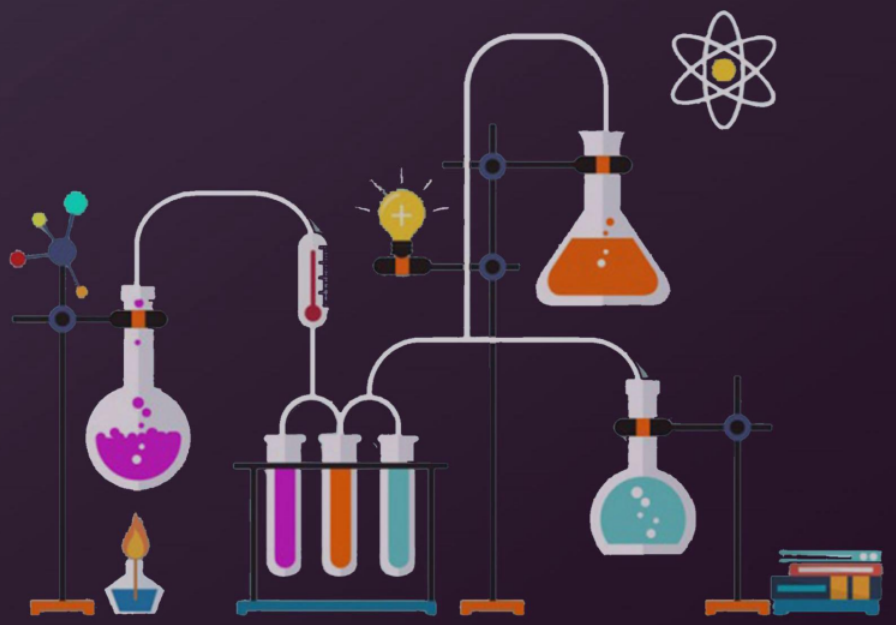
CHEMISTRY

Structural Isomerism



SHEET

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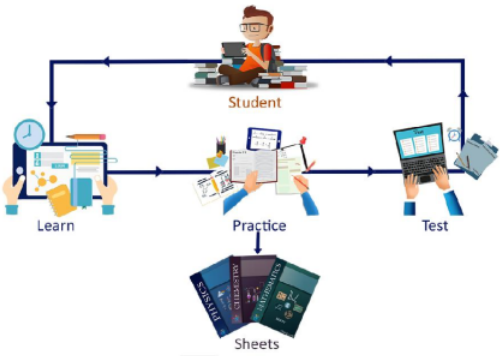
Contents

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2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 3
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

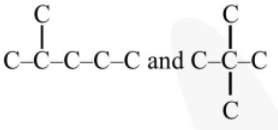
Sr. No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Isomerism and Structural Isomerism	1-4			
2	Ring Chain isomers, Functional Isomers and Metamers	5-11	1-2, 4-5, 8-9	1-9	1-2
3	Tautomerism	12		10-11	3, 7-9
4	Geometrical Isomerism - Part 1	13-15		12	
5	Geometrical Isomerism - Part 2	16-18	3, 6-7, 10-11	13-21	4, 6
6	Conformational Isomerism in Alkane - Part 1	19-20		22	
7	Conformational Isomerism in Alkane - Part 2				
8	Conformational Isomerism in Cycloalkane - Part 1	21			
9	Conformational Isomerism in Cycloalkane - Part 2			23-26	5
10	ACC	22-24	12-13	27-30	

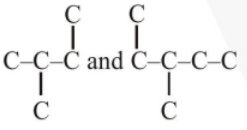
SINGLE CORRECT TYPE

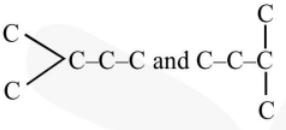
*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

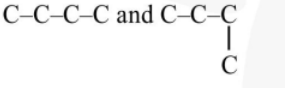
Introduction to Isomerism and Structural Isomerism

Q1. Which is the example of Chain isomerisation:

(A) 

(B) 

(C) 

(D) 

Q2. Which pair represents chain isomer :-

(A) CH_3CHCl_2 and $ClCH_2CH_2Cl$

(B) n-propyl alcohol and isopropyl alcohol

(C) 2-methyl-1 propanol and 2-Methyl-2-propanol

(D) 2-methyl butane and neopentane

Q3. What is relation between the given compounds $CH_3–CH=CH–CH_3$ and $CH_2=\underset{\displaystyle CH_3}{\underset{|}{C}}–CH_3$.

(A) Position isomers

(B) Chain isomers

(C) Homologues

(D) 2 & 3 both

Q4. Which of the following pairs is an example of position isomerism :-

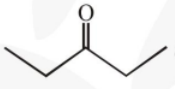
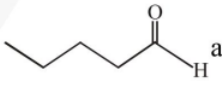
(A) $CH_3–CH_2–CH_2–CH_3$ and $CH_3–\underset{\displaystyle CH_3}{\underset{|}{CH}}–CH_3$

(B) $CH_3–CH_2–CH=CH_2$ and $CH_3–CH=CH–CH_3$

(C) $CH_3–CH_2OH$ and $CH_3–O–CH_3$

(D) $CH_3–\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}–CH_3$ and $CH_3–CH_2–CH_2–CH_2–CH_3$

Ring Chain Isomers, Functional Isomers and Metamers

Q5.  and  are :-

(A) chain isomers

(B) position isomers

(C) functional group isomers

(D) metamers

Q6. $CH_3–CH_2–\overset{\displaystyle O}{\overset{\|}{C}}–OH$ and $CH_3–\overset{\displaystyle O}{\overset{\|}{C}}–O–CH_3$ are :-

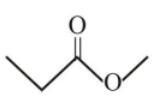
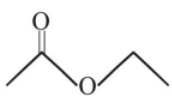
(A) Geometrical isomers

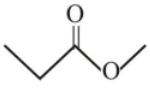
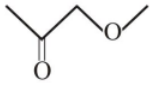
(B) Metamers

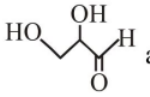
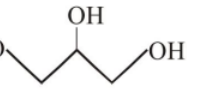
(C) Functional group isomers

(D) Stereo isomers

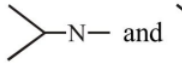
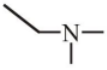
Q7. Which of the following pair of compound are functional isomers ?

(A)  and 

(B)  and 

(C)  and 

(D)  and 

Q8.  and  show which type of isomerism :-

(A) chain (B) position

(C) metamerism (D) functional group

Q9. The isomerism exhibited by alkyl cyanide and alkyl isocyanide is :-

(A) Functional (B) Positional

(C) Tautomerism (D) Metamerism

Q10. $BrCH_2–CH_2–CH=O$ and $CH_3–CH_2–\underset{\underset{Br}{|}}{C}=O$ are :-

(A) Functional isomers

(B) Position isomers

(C) Chain isomers

(D) Metamers

Q11. The relation between $CH_3–CH_2–CH_2–\overset{\overset{O}{||}}{C}–CH_3$ and $CH_3–CH_2–\overset{\overset{O}{||}}{C}–CH_2–CH_3$ is :-

(A) Stereo isomers

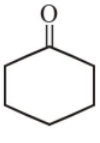
(B) Geometrical isomers

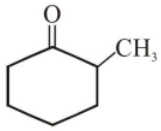
(C) Metamers

(D) Functional group isomers

Tautomerism

Q12. Which of the following will show tautomerism :-

(A) $CH_3–\overset{\overset{O}{||}}{C}–CH_3$ (B) 

(C)  (D) All of these

Geometrical Isomerism - Part 1

Q13. cis-but-2-ene and trans-but-2- ene will have:-

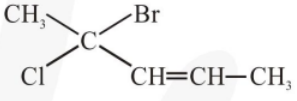
(A) same boiling point

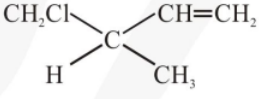
(B) same melting point

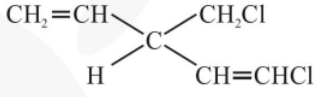
(C) same density

(D) same no. of atoms

Q14. One among the following will not show geometrical isomerism

(A) 

(B) 

(C) 

(D) $CH_3CH = CHCH_2CH_3$

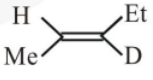
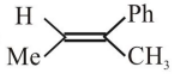
Q15. Fumaric acid and maleic acid are related to each other as :-

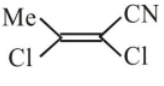
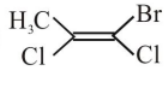
(A) Position-isomer (B) Stereo-isomer

(C) Chain-isomer (D) Tautomer

Geometrical Isomerism - Part 2

Q16. Which of the following represent Z from :-

(A)  (B) 

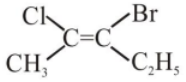
(C)  (D) 

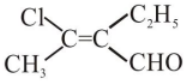
Q17. The number of cis-trans isomers possible for the following compound :-

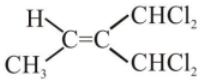


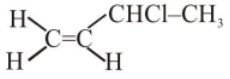
(A) 2 (B) 4 (C) 6 (D) 8

Q18. Select the E-isomer among the following :-

(A) 

(B) 

(C) 

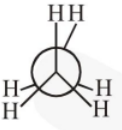
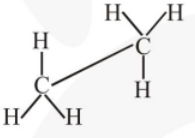
(D) 

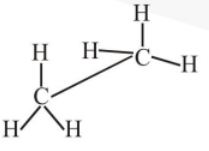
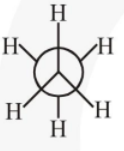
Conformational Isomerism in Alkane - Part 1

Q19. Total number of conformational isomers of ethane is :-

(A) 2 (B) 4 (C) 8 (D) ∞

Q20. Which are the staggered forms of ethane :-

(a)  (b) 

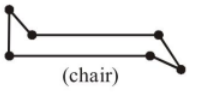
(c)  (d) 

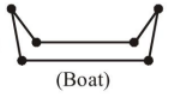
(A) (a) and (d) (B) (c) and (d)

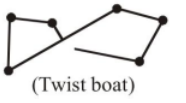
(C) (b) and (d) (D) (a) and (c)

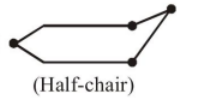
Conformational Isomerism in Cycloalkane - Part 1

Q21. Which of the following is the least stable conformer of cyclohexane ?

(A) 
(chair)

(B) 
(Boat)

(C) 
(Twist boat)

(D) 
(Half-chair)

ACC

Q22. The number of possible alkynes with molecular formula C_5H_8 is :-

(A) 2 (B) 3

(C) 4 (D) 5

Q23. How many structural isomeric primary amines are possible for the formula $C_4H_{11}N$?

(A) 2 (B) 3

(C) 4 (D) 5

Q24. C_7H_9N has how many isomeric forms that contain a benzene ring :-

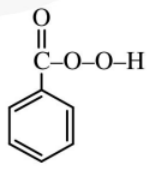
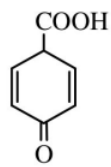
(A) 4 (B) 5

(C) 6 (D) 7

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1.  &  is a pair of :

(A) Ring chain isomer

(B) Position isomer

(C) Functional group isomer

(D) Structural Isomer

Q2. Which of the following represent correct matching ?

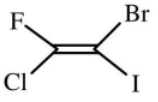
(A) $CH_3-\overset{\overset{O}{\|}}{C}-OH$ and $H-\overset{\overset{O}{\|}}{C}-O-CH_3$ — Metamer

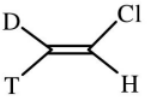
(B) $CH_3-CH_2-C\equiv CH$ and $CH_3-C\equiv C-CH_3$ — Position isomers

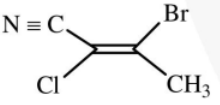
(C) $CH_3-CH_2-CH_2-NH_2$ and $CH_3-\underset{\underset{NH_2}{|}}{CH}-CH_3$ — Tautomers

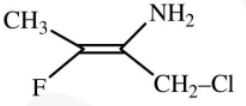
(D) CH_3-CH_2-OH and $(CH_3)_2O$ — Functional isomers

Q3. Identify 'E' configuration ?

(A) 

(B) 

(C) 

(D) 

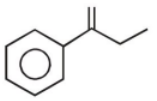
PARAGRAPH TYPE

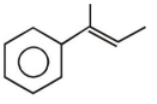
*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

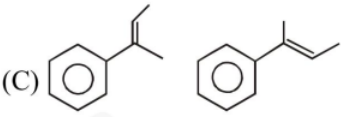
Paragraph for Questions 4 to 5

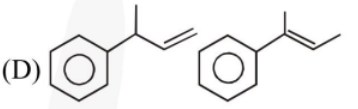
Structural isomerism for any compound may be chain, position, functional, metamerism & tautomerism.

Q4. Which are not positional isomer of each other-

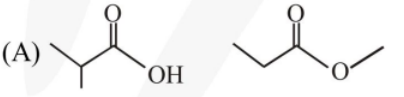
(A) 

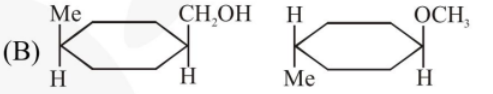
(B) 

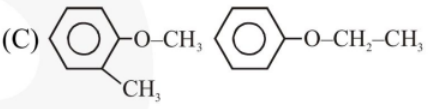
(C) 

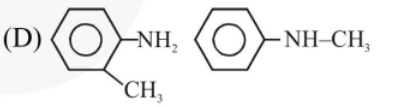
(D) 

Q5. Which are not functional isomers-

(A) 

(B) 

(C) 

(D) 

Paragraph for Questions 6 to 7

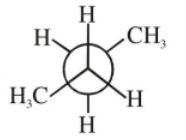
Compound containing sigma (σ) bond permits free rotation along σ bond due to symmetrical distribution of electrons around internuclear axis of sigma bond. The distribution of electrons does not disturbed due to rotation along sigma bond axis. The rotation about sigma bond result into different spatial arrangements of atoms groups in space which can change into one another. Such spatial arrangements of atoms / groups which can be converted into one another by rotation along a σ bond are called conformations or rotomers. Alkanes can thus have infinite number of conformations by rotation along carbon–carbon in single bond.

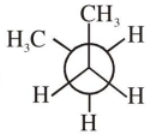
Q6. Compound among following which can form conformer :

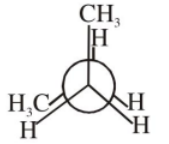
(A) CH_3–Cl (B) CH_3–OH

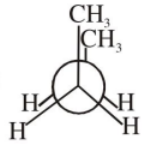
(C) CH_4 (D) H_2O

Q7. Which of the following conformation of butane is most stable due to rotation along C_2–C_3 bond :

(A) 

(B) 

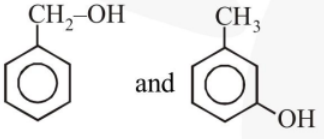
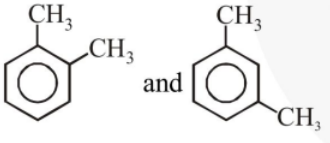
(C) 

(D) 

MATCHING LIST TYPE

Each question has TWO (02) ***matching lists: LIST I and LIST II****. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q8.

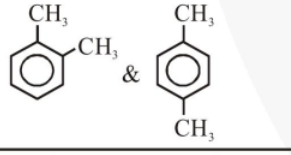
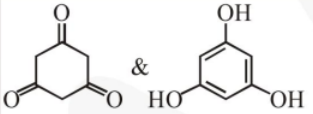
List-I (Compounds)		**List-II (Type of isomerism)**
(P)	$CH_3–CH_2–CH_2–\overset{\overset{O}{\|\|}}{C}–CH_3$ and $CH_3–CH_2–\overset{\overset{O}{\|\|}}{C}–CH_2–CH_3$	(1) Positional
(Q)		(2) Metamerism
(R)		(3) Functional
(S)	$CH_3–CH_2–\overset{\overset{CH_3}{\|}}{CH}–CH_3$ and $CH_3–CH_2–CH_2–CH_2–CH_3$	(4) Chain

	P	Q	R	S
(A)	1,2	3	1	4
(B)	3	2	4	1
(C)	2	3	4	1
(D)	4	1	2	3

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q9. Match column-I with column II and select correct answer from the codes given below the lists :

	Column-I		Column-II
(A)		(P)	Chain isomer
(B)		(Q)	Position isomer
(C)		(R)	Metamers
(D)		(S)	Tautomers

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q10. Identify total number of Geometrical isomers for the following compound :

Et-CH=CH-CH=CH-CH=CH-Me

Q11. Total number of structural isomers possible for C_4H_8.

Q12. How many structures of 1,4-disubstituted benzene is/are possible with molecular formula $C_8H_8O_3$ in which one –COOH group must be present :

Q13. How many structures are theoretically possible for formula C_5H_{10} only cyclic ?

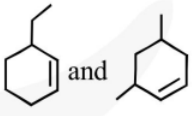
Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. 2-Pentanone and 3-methyl-2-butanone are a pair of ..... isomers :-

(A) Functional
(B) Chain
(C) Positional
(D) Stereo

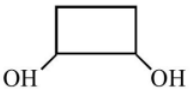
Q2.  and are :-

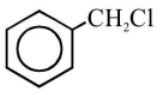
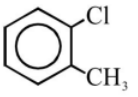
(A) Position isomers
(B) Chain isomers
(C) functional isomers
(D) metamers

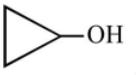
Q3. Minimum number of carbon atom required by a hydrocarbon alkane, alkene and alkyne to show chain isomerism respectively are :-

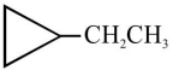
(A) 4, 4, 4 (B) 4, 4, 3
(C) 4, 5, 4 (D) 4, 4, 5

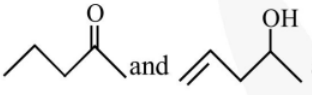
Q4. Which of the following pairs of compounds is a ring chain isomer :

(A) $CH_3–\underset{\underset{OH}{|}}{CH}–\underset{\underset{OH}{|}}{CH}–CH_3$ and 

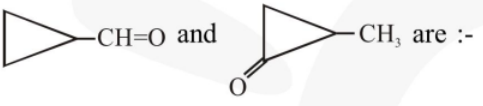
(B)  and 

(C) $CH_2=CH–CH_2–OH$ and 

(D)  and 

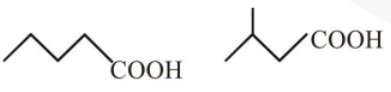
Q5.  are :-

(A) Tautomers
(B) Functional isomer
(C) Both (A) & (B)
(D) Metamers

Q6.  are :-

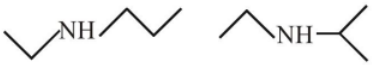
(A) Functional isomers
(B) Position isomers
(C) Geometrical isomers
(D) Optical isomers

Q7. Which of the following pair are metamers of each other :-

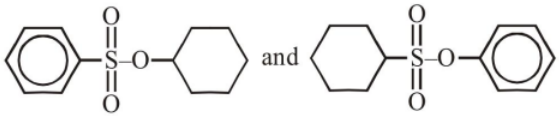
(A) 

(B) 

(C) 

(D) 

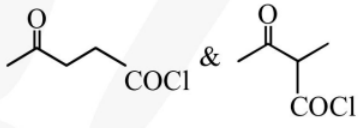
Q8. Given compound show which type of isomerism?

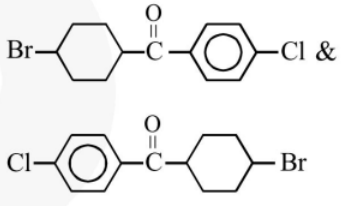


(A) Chain isomerism
(B) Positional isomerism
(C) Metamerism
(D) Functional group isomerism

Q9. Select incorrect pairs

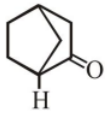
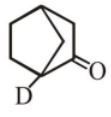
(A) $H_3C–CH_2–CH_2–NH–CH_3$
and $H_3C–\underset{\underset{CH_3}{|}}{CH}–NH–CH_3$
are positional isomers.

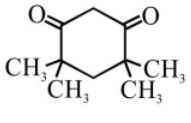
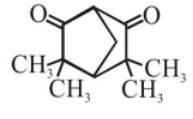
(B) 
are chain isomers.

(C) 
are metamers.

(D) Gauche form is the most stable form of ethane–1, 2–diol (ethylene glycol).

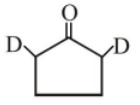
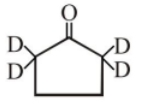
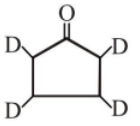
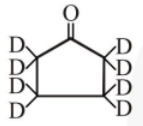
Q10. Which of the following will not tautomerise to enol :-

(A)  (B) 

(C)  (D) 

Q11.  $\xrightarrow{D_2O/\overset{\ominus}{O}D}$ Prolonged treatment of

(A) with $D_2O/\overset{\ominus}{O}D$ given :-

(A) 
(B) 
(C) 
(D) 

Q12. Which of the following can exhibit geometrical isomerism :-

(A) 1-Butene

(B) Methyl cyclopropane

(C) 2- methyl pent-2-ene

(D) 1, 2-dimethyl cyclopropane

Q13. According to CIP sequence rule, the correct arrangement of priority is :-

(A) –OH > $–CH_2OH$ > –CHO > – COOH

(B) –OH > –COOH > –CHO > $–CH_2OH$

(C) –COOH > –OH –CHO > $–CH_2OH$

(D) –COOH > –CHO > $–CH_2OH$ > –OH

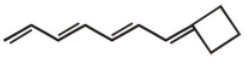
Q14. Total number of geometrical isomers in given compound :-

$CH_3–CH=CH–CH_2–CH=CH–CH_3$

(A) 4 (B) 3

(C) 5 (D) 6

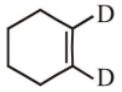
Q15. Total number of geometrical isomer of the following compound :

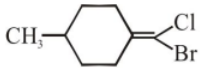


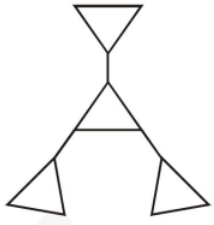
(A) 2 (B) 4

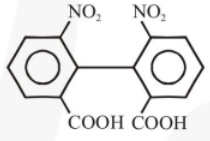
(C) 8 (D) 16

Q16. Which will show geometrical isomerism :-

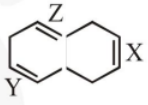
(A) 

(B) 

(C) 

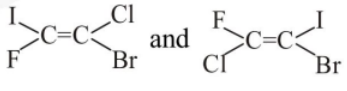
(D) 

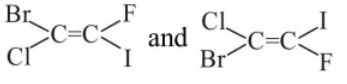
Q17. Types of geometrical isomerism shown at point X, Y and Z of the following compound respectively are –

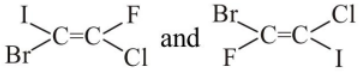


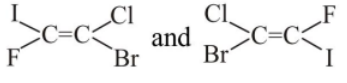
	X	Y	Z
(A)	cis	cis	trans
(B)	cis	trans	trans
(C)	trans	cis	cis
(D)	cis	trans	cis

Q18. Which of the following pairs of compounds are geometrical isomers :

(A) 

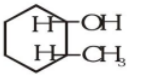
(B) 

(C) 

(D) 

Q19. Which among the following will show geometrical isomerism :

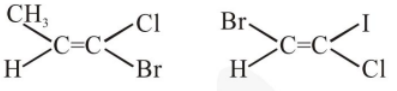
(A) 

(B) 

(C) C_6H_5–CH=N–OH

(D) All of these

Q20. Which of the following is true about A and B ?



(A) (B)

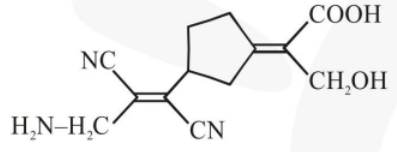
(A) (A) → Z; (B) → E

(B) (A) → E; (B) → E

(C) (A) → E; (B) → Z

(D) (A) → Z; (B) → Z

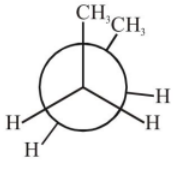
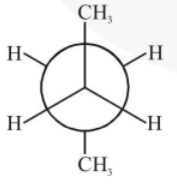
Q21. Assign double bond configurations to the following :-

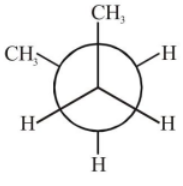
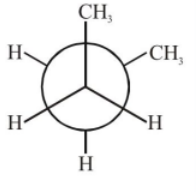


(A) E, Z (B) Z, E

(C) E, E (D) Z, Z

Q22. Which of the following is maximum stable conformation of butane :-

(A)  (B) 

(C)  (D) 

Q23. Increasing order of stability among the three main conformations (i.e., Eclipse, Anti, Gauche) of 2-fluoroethanol is :-

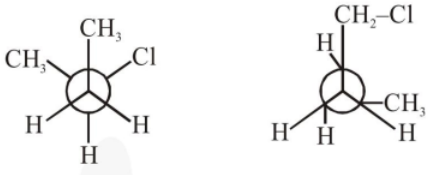
(A) Anti, Gauche, Eclipse

(B) Eclipse, Gauche, Anti

(C) Gauche, Eclipse, Anti

(D) Eclipse, Anti, Gauche

Q24. What is relation between a and b ?



(a) (b)

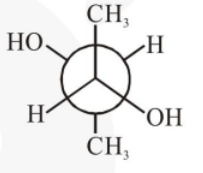
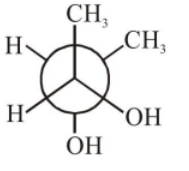
(A) (a) and (b) are conformatinal isomers

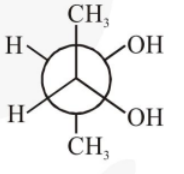
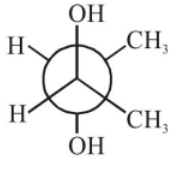
(B) (a) and (b) are geometrical isomers

(C) (a) and (b) are position isomers

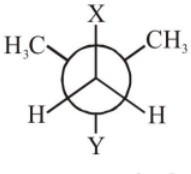
(D) (a) and (b) are chain isomers

Q25. Which of the following is the most stable conformer?

(A)  (B) 

(C)  (D) 

Q26. In the Newman projection for 2, 2-dimethylbutane :



X and Y can respectively be :-

(A) H and H (B) H and C_2H_5

(C) C_2H_5 and H (D) CH_3 and CH_3

(A) A and B (B) B and C

(C) B and D (D) C and D

Q27. How many isomers of $C_5H_{11}OH$ will be primary alcohols (only structure) :-

(A) 2 (B) 3

(C) 4 (D) 5

Q28. The total number of structural isomers of $C_4H_{10}O$ will be :-

(A) 4 (B) 5

(C) 6 (D) 7

Q29. How many structural isomers will be for C_5H_{10} ? (only cyclic)

(A) 3 (B) 4 (C) 5 (D) 6

Q30. Molecular formula $C_5H_{10}O$ can have structural isomers :-

(A) 6-Aldehyde, 4-ketone

(B) 4-Aldehyde, 3-ketone

(C) 3-Aldehyde, 3-ketone

(D) 5-Aldehyde, 2-ketone

Exercise - 2A — *Brain Booster*

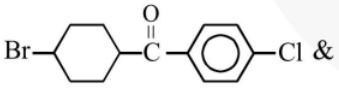
MORE THAN ONE CORRECT TYPE

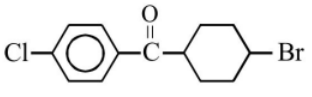
*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Select incorrect pair from the following :

(A) $H_3C–CH_2–CH_2–NH–CH_3$ & $H_3C–\underset{\displaystyle CH_3}{\underset{|}{CH}}–NH–CH_3$ are chain isomers

(B)  are positional isomers

(C)  &

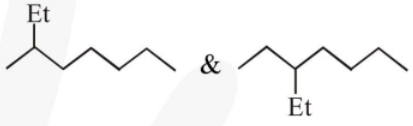


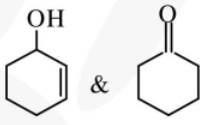
are metamers

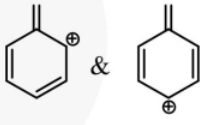
(D) $CH_3–CHO$ & 

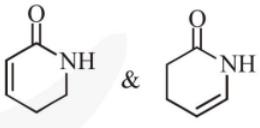
are ring chain isomers

Q2. Select the correct statement from the following:

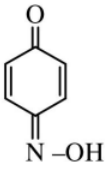
(A)  are positional isomers

(B)  are functional isomers

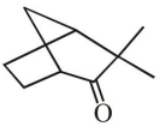
(C)  are positional isomers

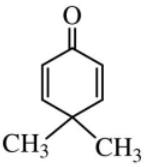
(D)  are positional isomers

Q3. Which of the following compound will not show tautomerism ?

(A) 

(B) $Me_3C–C\equiv N$

(C) 

(D) 

Q4. Number of geometrical isomers possible for the following compound :



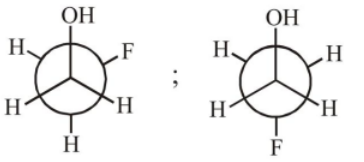
(A) 4

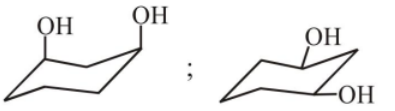
(B) 2

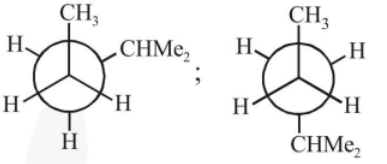
(C) no G.I. possible

(D) 8

Q5. In which case 2nd structure is less stable than Ist structure.

(A)  ;

(B)  ;

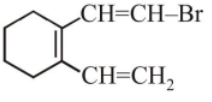
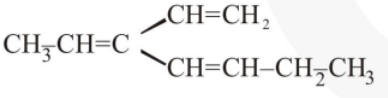
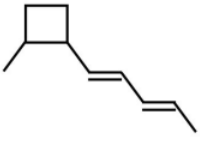
(C)  ;

(D)  ;

MATCHING LIST TYPE

Each question has TWO (02) ***matching lists: LIST I and LIST II****. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q6.

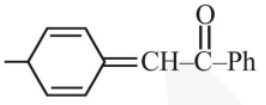
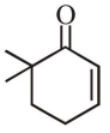
List - I (Compound)	**List - II (Number of geometrical isomers)**
(P) $CH_3–CH=CH–CH=CH–CH_3$	(1) 4
(Q) 	(2) 3
(R) 	(3) 8
(S) 	(4) 2

	P	Q	R	S
(A)	2	3	1	4
(B)	2	4	1	3
(C)	2	1	4	3
(D)	3	1	2	4

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q7.

Column-I (Compounds)	**Column-II (Number of hydrogen exchanged by deutarium in solution prdong treatment)**
(A) $Ph–CH_2–CH=CH–\overset{\overset{O}{\|}}{C}–CH_3$	(P) 5
(B) 	(Q) 9
(C) 	(R) 4
(D) $CH_3–CH_2–\overset{\overset{O}{\|}}{C}–CH_3$	(S) 3

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q8. How many cyclic structural isomers are possible for molecular formula C_4H_6 ?

Q9. How many structural isomers are possible of $C_4H_{11}N$.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
A	D	B	B	C	C	B	D	A	A
11	12	13	14	15	16	17	18	19	20
C	D	D	B	B	C	A	B	D	C
21	22	23	24						
D	B	C	B						

Ex-1A

1. C, D **2.** B,D **3.** B,C,D **4.** C **5.** C

6. B **7.** A **8.** A

9. A **10.** 8 **11.** 5 **12.** 003 **13.** 5

Ex-2									
1	2	3	4	5	6	7	8	9	10
B	B	D	C	B	A	B	C	C	D
11	12	13	14	15	16	17	18	19	20
B	D	B	B	B	C	A	D	D	C
21	22	23	24	25	26	27	28	29	30
C	B	D	C	C	C	C	D	C	B

Ex-2A

1. A,B,C **2.** B,D **3.** B,C,D **4.** B **5.** A,B,D

6. B **7.** (A)→Q ; (B)→R ; (C)→S ; (D)→P **8.** 5

9. 8

-: Important Notes & Formulas :-